

REQUIREMENT DEFINITION PROPOSAL (RDP)

RANGANADI-BETA

Smart City Citizen Engagement & Service Management Platform

1. Project Overview

RANGANADI-BETA is a comprehensive Web and Mobile-based Smart City platform designed to connect citizens, city administrators, departments, and field teams through a single digital ecosystem.

The platform aims to improve citizen engagement, service delivery, issue resolution, public communication, and data-driven decision-making through integrated digital services and GIS-enabled features.

2. Project Vision

To create a citizen-centric digital platform that enhances communication between city authorities and residents while improving operational efficiency, transparency, and accessibility of public services.

3. Core Users

Citizens

- Access city services
- Submit complaints and requests
- Participate in surveys
- Provide feedback
- Receive announcements and alerts

Survey Teams

- Conduct field surveys
- Collect geo-tagged data
- Upload field observations

Department Officers

- Manage complaints and tickets
- Monitor service requests

- Update resolution status

City Administrators

- Monitor city operations
 - Manage departments and users
 - Analyze reports and performance metrics
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4. System Modules

Module 1: Citizen Portal

Features

- Citizen Registration & Login
 - Personal Dashboard
 - Service Requests
 - Complaint Submission
 - Complaint Tracking
 - Citizen Feedback
 - Notifications & Alerts
 - Announcements
 - Public Information Services
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Module 2: Survey Management

Features

- Dynamic Survey Forms
 - Geo-Tagged Survey Collection
 - Image Upload
 - Offline Survey Capability
 - Survey Assignment
 - Survey Monitoring Dashboard
 - Survey Analytics
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Module 3: Complaint Management

Features

- Complaint Registration
- Category-Based Complaints

- GPS Location Capture
- Photo & Video Upload
- Complaint Assignment
- Complaint Tracking
- Resolution Management

Status Flow

- Submitted
- Assigned
- Under Review
- In Progress
- Resolved
- Closed

Module 4: Ticket Management System

Features

- Automatic Ticket Generation
- Unique Ticket IDs
- Department Assignment
- Escalation Workflow
- SLA Monitoring
- Ticket Tracking
- Resolution History

Module 5: Reports & Analytics

Features

- Daily Reports
- Weekly Reports
- Monthly Reports
- Department Performance Reports
- Complaint Analytics
- Survey Analytics
- Citizen Engagement Metrics

Export Formats

- PDF
- Excel
- CSV

Module 6: Announcement & Notification System

Features

- Public Announcements
 - Emergency Alerts
 - Service Notifications
 - Event Notifications
 - Push Notifications
 - In-App Notifications
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Module 7: Citizen Feedback System

Features

- Service Rating
 - Citizen Suggestions
 - Satisfaction Surveys
 - Department Feedback
 - Feedback Analytics
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5. GIS-Based Public Services

Toilet Locator

Features

- Nearby Toilet Search
 - GPS Navigation
 - Facility Details
 - Real-Time Location Display
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Office Locator

Features

- Government Office Search
 - Department Filtering
 - GPS Navigation
 - Office Details
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Officials Directory

Features

- Department-Wise Contacts
 - Emergency Contacts
 - Click-to-Call Support
 - Email Directory
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6. Dashboard Overview

Citizen Dashboard

- My Complaints
- Service Requests
- Notifications
- Feedback History
- Survey Participation

Department Dashboard

- Assigned Tickets
- Pending Cases
- Resolution Performance
- Service Metrics

Admin Dashboard

- Total Citizens
 - Total Complaints
 - Open Tickets
 - Closed Tickets
 - Survey Statistics
 - Citizen Satisfaction Score
 - Department Performance
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7. Technology Stack

Frontend

- React.js (Web Portal)
- Flutter (Mobile Application)

Backend

- Node.js
- Express.js

Database

- PostgreSQL / MySQL

GIS & Mapping

- Google Maps API

Authentication

- Mobile OTP Login
- JWT Authentication

Notifications

- Firebase Cloud Messaging (FCM)

Hosting Infrastructure

- AWS Cloud
- Hostinger VPS
- Azure (Optional)

8. Security & Compliance

- SSL Encryption
- Role-Based Access Control (RBAC)
- Secure API Authentication
- Activity Logs
- Data Backup & Recovery
- User Access Audit Trails

9. Expected Deliverables

- Citizen Mobile Application
- Citizen Web Portal
- Administrative Dashboard
- Survey Management System
- Complaint Management System
- Ticket Management System

- Reports & Analytics Portal
 - Notification Management System
 - GIS-Based Public Facility Locator
 - Officials Directory
 - API Documentation
 - Deployment & Training Documentation
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10. Expected Outcomes

- Enhanced Citizen Engagement
 - Faster Issue Resolution
 - Improved Transparency
 - Better City Service Monitoring
 - Centralized Information Access
 - Data-Driven Governance
 - Improved Citizen Satisfaction
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Project Name: RANGANADI-BETA

Project Type: Smart City Citizen Engagement & Service Management Platform

Version: 1.0